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Artificial Intelligence of Things as a Foundation for Agentic AI Systems: Architectures, Applications, and Challenges

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Keywords

artificial intelligence of things (aiot), autonomous agents, computing and processing, edge computing, intelligent systems

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Index Terms—Artificial Intelligence of Things (AIoT), Autonomous Agents, Edge Computing, Intelligent Systems

I. INTRODUCTION

Artificial intelligence (AI) has progressed through successive paradigms, from predictive AI that primarily performs pattern recognition and inference, to generative AI that can synthesize text, images, and code at unprecedented scale. The recent emergence of Agentic AI marks a further shift from producing responses to pursuing goals: agentic systems continually perceive context, plan over long horizons, invoke external tools, and execute actions in iterative feedback loops.

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This transition is catalyzed by large language models (LLMs), whose web-scale pretraining endows broad world knowledge and instruction-following abilities. As a result, LLM-based autonomous agents have rapidly become a central research topic, with surveys proposing unified frameworks that integrate planning, memory, tool use, and evaluation methods [1]. Tool learning has formalized how foundation models decompose complex objectives into subtasks and dynamically select tools to complete them more reliably and efficiently [2]. Beyond general automation, agentic workflows have already demonstrated transformative potential in high-stakes domains, such as medicine [3] and scientific discovery, where systems can autonomously design and execute experiments by orchestrating tools and laboratory automation [4]. To position this survey, Table I compares it with representative related surveys on LLM-based agents, tool learning, embodied robotics, and AIoT/edge security. Existing works typically focus on one or two dimensions, while missing an end-to-end view that jointly covers agentic capability, embodiment, AIoT infrastructure, edge-cloud systems, and security/trust.

Despite this momentum, many agentic systems remain fundamentally disembodied. They operate in digital environments with simplified observations, clean APIs, and low-cost trial-and-error, which diverges from human learning that is grounded in physical interaction. When deployed beyond controlled software settings, disembodied agents face persistent limitations: partial observability, noisy sensing, delayed and sparse feedback, uncertain actuation, and stringent safety and resource constraints. These gaps become most apparent in robotics, where successful autonomy requires coupling high-level reasoning with low-level sensorimotor control. Recent efforts toward embodied intelligence illustrate both the promise and the challenge: embodied LLM-enabled robots can complete long-horizon tasks in unpredictable environments by combining retrieval-augmented knowledge with force/vision feedback, but they also highlight the need for robust real-world perception-action grounding [5]. Systematic reviews of foundation models for robot learning likewise emphasize unresolved issues such as generalization under dynamics shifts, hardware-software decoupling, and safe interaction in human-centric spaces [6].

The Artificial Intelligence of Things (AIoT) offers a practical way to bridge this disembodiment by embedding agentic intelligence into distributed sensing, communication, computation, and actuation infrastructures. AIoT enables continuous context acquisition from heterogeneous devices, on-site decision-making at the edge, and closed-loop control

TABLE I
COMPARISON BETWEEN THIS SURVEY AND EXISTING/RELATED SURVEYS.

Work	Agentic AI	Tool Use	Embodiment	AIoT/IoT	Edge/Cloud Sys.	Security/Trust	Eval./Data
Wang et al. [1]	✓	✓	~	×	~	~	✓
Qin et al. [2]	~	✓	×	×	~	✓	~
Xiao et al. [3]	~	~	×	×	~	~	✓
Boiko et al. [4]	✓	✓	×	×	~	~	~
Mon et al. [5]	✓	✓	✓	×	~	~	~
Xiao et al. [6]	✓	✓	✓	×	~	~	✓
De Caro et al. [7]	×	×	×	✓	×	✓	✓
Somvanshi et al. [8]	×	×	~	✓	✓	~	✓
Le et al. [9]	×	×	×	✓	✓	✓	~
Rana et al. [10]	×	×	×	✓	~	✓	~
This survey (AIoT-enabled Agentic AI)	✓	✓	✓	✓	✓	✓	✓

✓: explicitly covered; ~: partially/implicitly related; ×: not a focus.

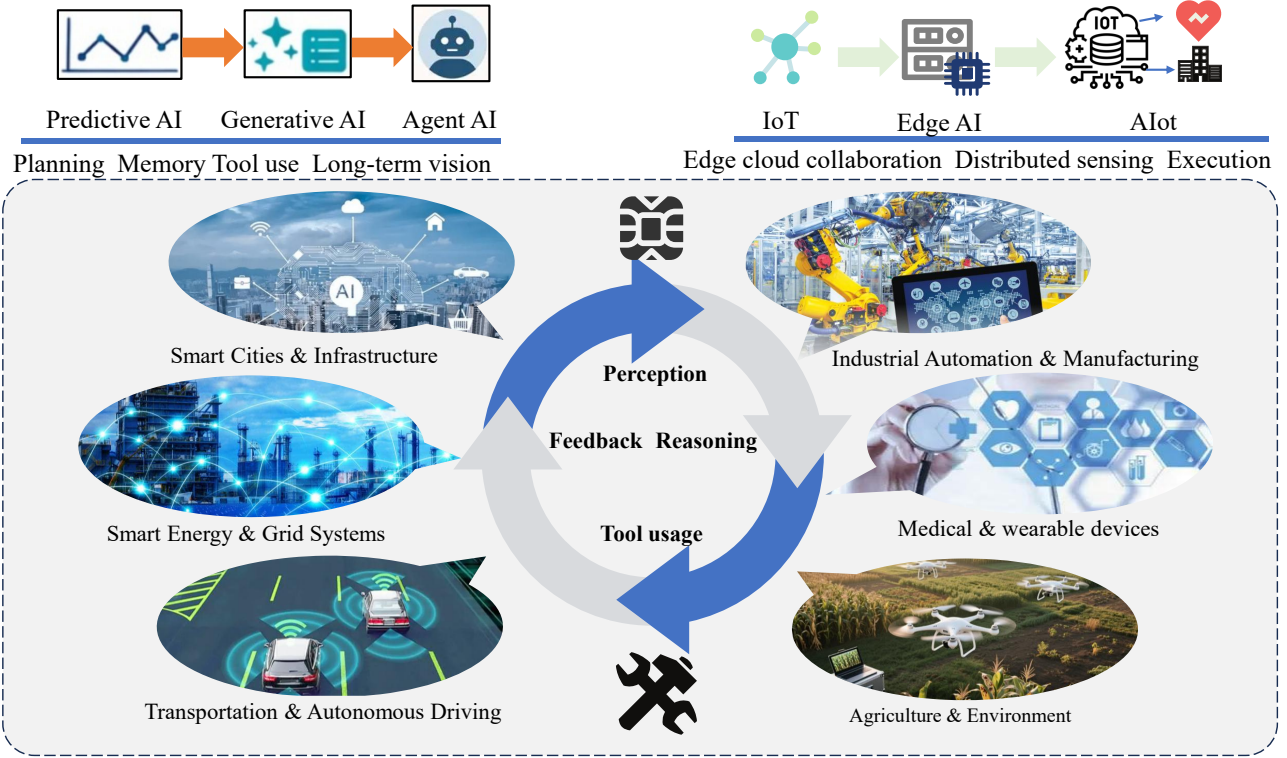


Fig. 1. Conceptual architecture of AIoT-enabled Agentic AI Systems. The figure illustrates a unified view of agentic AI empowered by AIoT, highlighting the parallel evolution of AI paradigms from predictive to generative and agentic intelligence, and of computing infrastructures from IoT to edge AI and AIoT. The central module depicts an agentic closed-loop consisting of sensing, reasoning/planning, tool use, action/actuation, and feedback. Surrounding application domains demonstrate representative AIoT-driven scenarios.

of physical processes. Edge intelligence is especially important for real-world autonomy, as latency, bandwidth, and privacy constraints often prevent cloud-only reasoning. However, edge deployments must also remain trustworthy under heterogeneous and resource-limited conditions. This drives the need for lightweight and efficient on-device learning (e.g., TinyML/TinyDL) to bring perception and control closer to sensors and actuators [8]. At scale, AIoT autonomy increasingly relies on distributed learning and inference across devices and edge nodes to coordinate services such as security, localization, and industrial control in emerging networks [9]. Additionally, realistic and reproducible AIoT agent evaluation

requires representative datasets and testbeds, particularly for security and traffic intelligence [7]. Finally, since agentic autonomy increases the attack surface through tool use and actuation, AIoT systems must include lightweight cryptographic protections appropriate for constrained devices [10].

This survey examines the emerging convergence of Agentic AI and the AIoT, aiming to clarify how goal-driven agents can be grounded in physical environments through distributed sensing, communication, edge–cloud computing, and actuation. We move beyond model-centric discussions of LLM agents and instead adopt a systems and architecture perspective: we connect the evolution from predictive and generative

AI to agentic intelligence with the parallel evolution from cloud-centric IoT to edge-native AIIoT, and analyze how this co-evolution enables real-world autonomy through closed-loop perception–decision–action pipelines. Building on this scope, the survey makes the following contributions:

- We synthesize the evolution toward agency and the shift from IoT to AIIoT, and articulate the foundational synergy between Agentic AI and AIIoT. We also provide a structured taxonomy of deployment topologies (centralized cloud-centric, distributed edge-native, and hybrid cloud–edge) and coordination models (hierarchical, mesh/swarm, and hybrid), highlighting key trade-offs in latency, privacy, resilience, and scalability.
- We formalize AIIoT-enabled agent interaction as cognitive closed loops, encompassing intent-based semantic interaction and real-time feedback with adaptive correction, and show how autonomy emerges from continuous coupling among sensing, planning, and actuation. We organize enabling technologies into four pillars: edge intelligence and efficient learning; communication and coordination mechanisms; trust, security, and privacy enablers; and digital twins and simulation platforms. We further link each pillar to its functional role in the agentic loop.
- We survey representative vertical domains (smart cities, industrial manufacturing, healthcare/wearables, and smart energy/grids) to illustrate how agentic capabilities are instantiated through AIIoT infrastructures. We also identify open problems related to safety, robustness, trust, resource-aware autonomy, societal concerns, and benchmarking/standardization, and outline actionable future directions aligned with real deployment constraints.

To clarify the overall landscape of AIIoT-enabled agentic intelligence and its key components, Fig. 1 provides a conceptual overview of AIIoT-enabled Agentic AI Systems. The figure integrates the parallel evolution of AI paradigms and intelligent infrastructures, and summarizes the closed-loop operational logic, representative application domains, and core enabling technologies.

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II. BACKGROUND

The convergence of AI and the Internet of Things (IoT) has undergone a transformative journey, shifting from simple data collection to autonomous, goal-oriented reasoning [11]. This evolution is marked by the transition from passive sensing to active agency, where systems not only predict or create but also act within the physical world. The transition from disembodied intelligence to agentic systems is supported by a dual-track technological progression shown in Fig. 2. While AI has shifted from forecasting patterns to goal-oriented reasoning, IoT has evolved from simple data collection to an intelligent substrate capable of supporting on-device inference and local autonomy.

A. Evolution of Artificial Intelligence Towards Agency

The trajectory of AI can be viewed through three distinct paradigms, each increasing in complexity and autonomy:

- *Predictive AI* is the foundational stage that focuses on pattern recognition and forecasting based on historical data. It answers the question “what is likely to happen” but lacks the ability to generate new content or take independent action [12].
- *Generative AI* represents a significant advancement in capability. It uses Large Language Models (LLMs) to generate new data, content, or code [13]. Although highly creative, it remains largely “disembodied,” operating within digital silos without a direct feedback loop from the physical environment.
- *Agentic AI* is the current frontier, involving AI systems designed with “agents”, the ability to perceive their environment, reason through complex tasks, and execute actions to achieve specific goals [14]. Unlike its predecessors, Agentic AI functions as an autonomous entity capable of long-term planning and tool use.

Finally, Table II highlights the fundamental shifts from the predictive models of the past to the autonomous agentic systems of the future.

B. Evolution from IoT to AIIoT

Parallel to the growth of AI, the infrastructure of connected devices has evolved to support increasingly heavy computational demands. (i) *Conventional IoT architectures* were characterized by a cloud-centric model in which simple sensors collected data and transmitted it to a central server for processing [15]. This model experienced high latency and significant bandwidth pressure. (ii) *Edge Intelligence with AIIoT* moves intelligence toward the edge to address latency and privacy concerns [16]. AIIoT integrates AI directly into the IoT ecosystem, enabling devices to perform local inference and continuous learning. This shift allows for reflexive decisions at the device level while reserving the cloud for deep reasoning.

C. Foundational Synergy Between Agentic AI and AIIoT

The synergy between Agentic AI and AIIoT is symbiotic. Agentic AI serves as the brain for autonomous decision-making [17], while AIIoT acts as the body or substrate required for real-world interaction. AIIoT transforms the environment into a perceivable and controllable space for the agent, supplying the real-time, multimodal data streams necessary for the agent to ground its reasoning in physical reality [18].

Limitations of Current Paradigms: Although the synergy between Agentic AI and AIIoT is clear in theory, current practical implementations face several systemic limitations. If deployments are executed, it may exhibit brittle autonomy, performing well only in controlled or simulated settings but failing unpredictably in real-world environments. Additionally, the cost and complexity of retrofitting existing infrastructure with the necessary sensing and actuation capabilities are often underestimated, hindering large-scale adoption. Even systems labeled as edge-intelligent often remain heavily reliant on cloud resources for core decision-making, contradicting the promise of true edge autonomy. Finally, the lack of interoperability among different vendors and platforms limits the vision

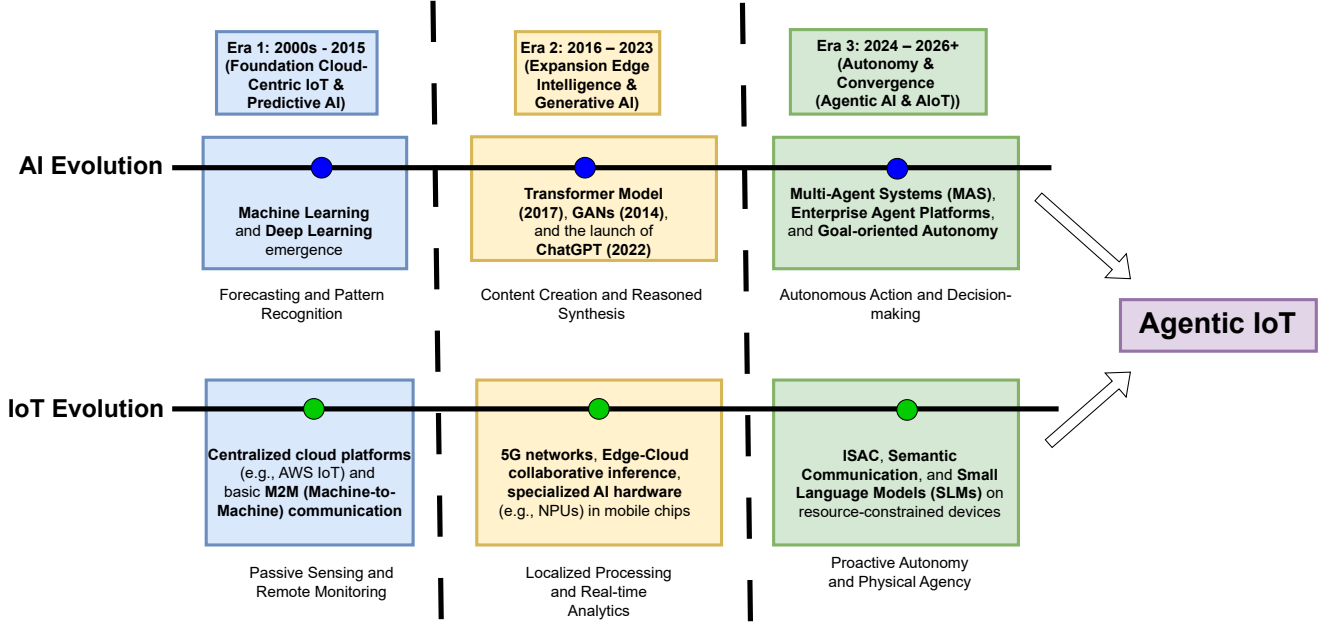


Fig. 2. The Evolutionary Roadmap: The convergence of the AI paradigm shift (Predictive $\rightarrow$ Generative $\rightarrow$ Agentic) with the IoT infrastructure shift (Cloud $\rightarrow$ Edge $\rightarrow$ AIoT)

TABLE II
COMPARATIVE ANALYSIS OF AI PARADIGMS IN THE AIOT LANDSCAPE

Paradigm	Key Capability	Data Requirement	Interaction Model	Limitation	Role of AIoT
Predictive AI	Pattern recognition & forecasting [12]	Structured historical data	Passive / Query-based	Lacks generative or autonomous action	Provides raw sensor data for model training
Generative AI	Content creation [13]	Massive unstructured datasets	Prompt-based / Reactive	Often "disembodied" and lacks physical grounding	Provides rich environmental context
Agentic AI	Autonomous reasoning & goal execution [14]	Real-time multi-modal feedback	Proactive & goal-oriented	High compute & trust/safety risks	Acts as the "body" for physical agency

of collaborative multi-agent ecosystems. These practical barriers highlight that future architectural designs must prioritize not only performance but also deployability and robustness in real-world conditions.

III. ARCHITECTURAL BLUEPRINT FOR AIOT-ENABLED AGENTIC SYSTEMS

To provide a comprehensive understanding of the structural evolution in Agentic AIoT, we present a unified taxonomy in Fig. 3. This visual framework synthesizes the architectural design along three critical dimensions: the physical placement of the cognitive brain, the organizational logic for agent coordination, and the microscopic mechanisms for interaction and control. The following subsections detail these paradigms respectively.

A. Centralized, Distributed, and Hybrid Architectures

The deployment of Agentic AI within the IoT ecosystem requires a fundamental re-evaluation of topological structures. Unlike traditional IoT systems that focus on data aggregation, Agentic AIoT architectures must support distributed inference,

continuous learning, and autonomous execution. These architectures can be categorized into three distinct paradigms based on the physical placement of the Agentic Brain.

1) *Centralized Cloud-Centric Architecture*: In this paradigm, the cognitive core of the agent, typically a LLM with massive parameters, resides entirely in the cloud. IoT devices primarily serve as sensory extensions and execution terminals, transmitting raw multimodal data to central servers for processing. The main advantage of this architecture is its ability to leverage virtually unlimited computational resources, allowing agents to perform complex reasoning and long-horizon planning using foundation models. However, as evidenced by the evolution of AIoT [19], this model suffers from significant latency bottlenecks and bandwidth pressure, rendering it unsuitable for real-time control tasks that demand millisecond-level responsiveness. Furthermore, the transmission of sensitive raw data (such as healthcare metrics or home audio) to the cloud raises severe privacy concerns, thereby limiting the deployment of centralized agents in sensitive domains.

2) *Distributed Edge-Native Architecture*: To address the latency and privacy shortcomings of cloud-centric models, distributed architectures push the agent's inference capabilities

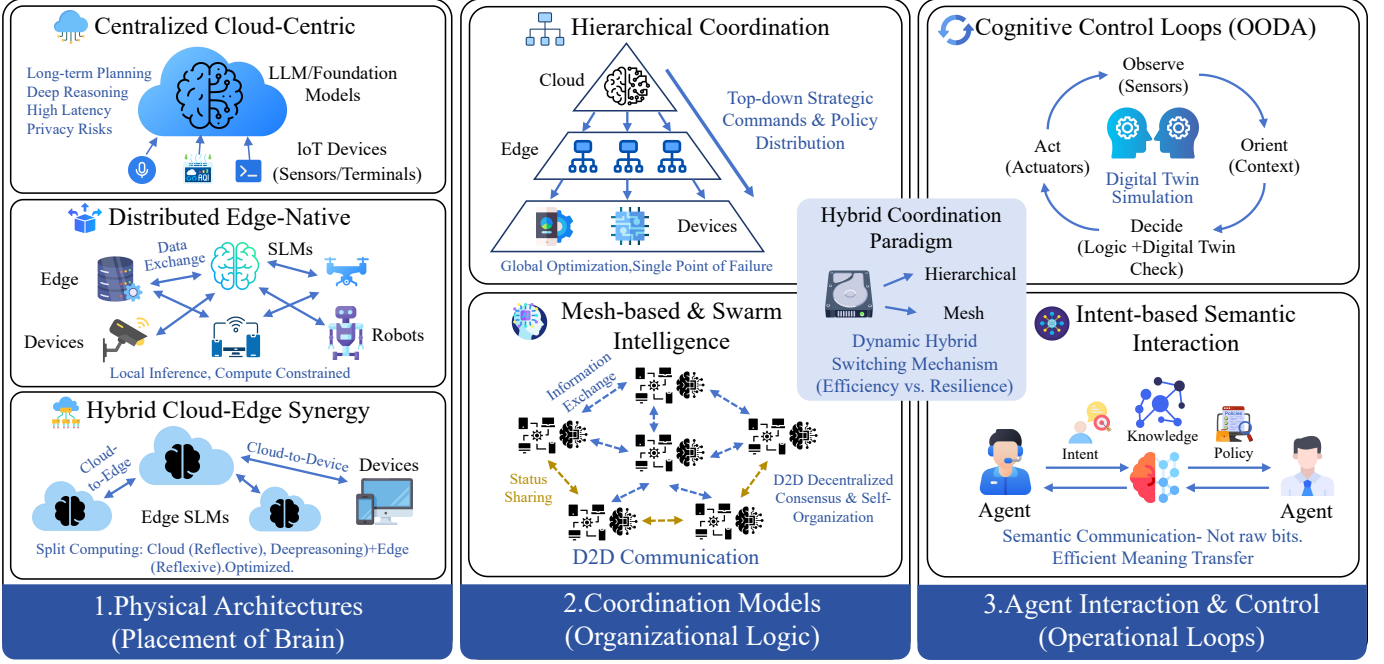


Fig. 3. Taxonomy of AIoT-enabled Agentic Architectures

directly to the network edge or on-device. By integrating AI directly into the IoT ecosystem, this approach empowers devices to perform local inference and decision-making without reliance on constant connectivity. Facilitated by advancements in model quantization and the emergence of Small Language Models (SLMs), edge-native agents can execute reflexive actions locally [20]. This architecture is essential for safety-critical applications, such as autonomous vehicles or industrial robotics, where network instability cannot be allowed to compromise safety. However, the constraints of edge hardware inevitably limit the depth of reasoning and the size of the context window, potentially restricting the agent's ability to handle highly ambiguous or novel situations compared to their cloud-based counterparts.

3) *Hybrid Cloud-Edge Synergy*: The most promising architectural blueprint for Agentic AIoT is the hybrid or split computing model. This hierarchical approach creates a symbiotic relationship where the brain is dynamically distributed across the device-edge-cloud continuum [21]. The device layer acts as the agent's reflexive system, handling sensing, shallow feature extraction, and immediate, low-latency execution tasks. The cloud layer serves as the agent's reflective system, responsible for in-depth reasoning, intensive model training, and long-term strategic planning. By dynamically partitioning the agent's workload, this hybrid architecture maximizes inference accuracy while ensuring real-time responsiveness. It allows the system to maintain high-level intelligence via the cloud while retaining the robustness and privacy benefits of edge execution, effectively resolving the dichotomy between model capability and deployment feasibility.

B. Hierarchical and Mesh-based Coordination Models

1) *Hierarchical Coordination Model*: The hierarchical coordination model emulates the strict chain-of-command structure characteristic of human organizations, making it particularly suitable for scenarios demanding rigorous global optimization, such as smart city or power grid management. In this architecture, the top-level Agent residing in the cloud assumes the role of a strategic commander, responsible for formulating macro-strategies and global resource scheduling. It leverages a comprehensive global view to establish long-term objectives (e.g., optimizing city-wide traffic flow) and decomposes these high-level intents into specific sub-tasks for downstream distribution. Meanwhile, middle and bottom-tier Agents situated at the edge or on devices are tasked with local tactical execution and micro-control; while they receive directives from superiors, they retain a degree of execution discretion to rapidly respond to local emergent conditions. The primary advantage of this top-down planning approach lies in its ability to ensure the consistency of overall system goals, thereby achieving optimal solutions through global coordination. However, this dependency also inherits the inherent fragility of hierarchical structures; should the top-level node fail or critical communication links be severed, lower-level Agents deprived of global guidance may deteriorate into suboptimal states due to information silos, potentially leading to system paralysis.

2) *Mesh-based and Swarm Intelligence Model*: To address the uncertainties inherent in dynamic environments and mitigate the single-point-of-failure risks associated with hierarchical structures, the mesh-based coordination model adopts a decentralized self-organization logic, marking a fundamental shift in control paradigms from centralized models to distributed Swarm Intelligence. In this architecture, the system

eschews reliance on centralized command nodes; instead, each agent exchanges states, intents, and policies directly with its neighbors via device-to-device communication. By leveraging advanced algorithms such as Multi-Agent Reinforcement Learning (MARL) [22], agents can dynamically adapt their strategies based on observations of the local environment and the behavior of neighboring nodes, enabling complex global intelligent behaviors to emerge from local interactions in the absence of a central controller. This mechanism is particularly well-suited for highly dynamic scenarios, such as UAV swarms or disaster relief operations. Even if a subset of nodes is damaged or destroyed, the remaining agents can autonomously reorganize connectivity via the mesh network to sustain mission execution, thereby significantly enhancing system robustness and fault tolerance.

3) *Hybrid Coordination Paradigm*: In practical complex industrial applications and critical infrastructure, to balance global efficiency and survivability, architectural designs are trending towards a hybrid coordination paradigm that integrates hierarchical and mesh-based approaches. This hybrid architecture allows the system to accept global optimization from hierarchical management to maximize operational efficiency under normal conditions characterized by stable network connectivity. Conversely, in extreme scenarios such as cyberattacks or natural disasters resulting in communication interruptions, Edge Agents can automatically switch to an autonomous coordination mode via mesh protocols. Through this dynamic switching mechanism, the AIoT system not only maintains the orderliness of macro-scheduling but also possesses the resilience required for independent survival in hostile environments, thereby ensuring the safety baseline of physical operations.

C. Agent Interaction and Control Loops

1) *Cognitive Control Loops*: To achieve true physical autonomy, the operational logic of AIoT Agents must transcend the traditional linear workflow of sense-transmit-process and transition toward a cognitive closed-loop based on the observe-orient-decide-act cycle. In this architecture, the Agent is no longer a mere data conduit but an entity possessing complete cognitive capabilities. First, the Agent observes the physical environment through multimodal sensors, utilizing on-board models to extract features from heterogeneous data. Subsequently, it incorporates environmental context to orient and reason, assessing the discrepancy between the current state and the target state. Following this, it decides on the optimal sequence of actions based on logical reasoning. Crucially, this architecture integrates digital twin technology to conduct Look-ahead simulation. Before executing any physical action, the Agent simulates the consequences in the virtual space, thereby selecting the optimal and safest course of action. This closed-loop mechanism empowers the system to shift from passive response to proactive planning, ensuring operational safety in low-error-tolerance scenarios such as industrial control or surgical robots.

2) *Intent-based Semantic Interaction*: At the interaction level, Agentic AIoT introduces the semantic communication

paradigm, fundamentally transforming the mode of information exchange between nodes. Distinct from traditional communication theory, which prioritizes the accuracy of bit transmission, the interaction between Agents focuses on the efficient transfer of Intents, Knowledge, and Policies. By leveraging the encoding capabilities of Generative AI, the transmitting Agent extracts the semantic features of the information, while the receiver performs reconstruction. This approach significantly enhances communication efficiency, enabling Agents to convey complex task instructions with minimal data volume. This is particularly critical for bandwidth-constrained or unstable edge environments, as it not only reduces communication latency but also eliminates interpretative ambiguity among heterogeneous devices, ensuring the precise transmission of multi-agent collaboration directives.

3) *Real-time Feedback and Adaptive Correction*: To ensure the robustness of physical operations, the control loop must integrate real-time feedback and self-correction mechanisms. The Agent must not merely execute actions but also maintain a continuous feedback loop, monitoring the physical world's response to its behavior in real time. When actual results deviate from the expected model, or when runtime anomalies are detected, the Agent can trigger an immediate re-planning process. By leveraging reflexive reasoning capabilities at the edge, the Agent can rapidly rectify errors locally without waiting for high-latency cloud instructions. This adaptive control loop ensures that the system maintains safe behavior even when confronted with adversarial interference or sensor noise, thereby effectively preventing physical accidents caused by perception spoofing or execution errors.

IV. KEY ENABLING TECHNOLOGIES

The realization of Agentic AIoT systems is not simply a matter of stacking technologies but requires technological breakthroughs across multiple dimensions, including computing, communications, security, and simulation. This chapter explores the core enabling technologies that support the deployment of agents from the cloud to the edge, enabling efficient collaboration and trusted interaction.

A. Edge Intelligence and Efficient Learning

To endow IoT devices with genuine "Autonomy", agents must be able to perform inference and even continuous learning at the edge, rather than relying solely on cloud connectivity. Since Agentic AI, particularly LLM-based agents, typically involves massive parameter counts, deploying them on resource-constrained IoT devices has become a major challenge.

First, edge-side model optimization and lightweighting are critical for addressing computational bottlenecks. Traditional deep neural network compression techniques are evolving into specialized technologies tailored for Large Language Models (LLMs) [23]. Advancements in quantization technology are making it increasingly feasible to run agents at the 7B-parameter scale on edge devices such as Raspberry Pis or high-end mobile chips. Meanwhile, the rise of SLMs has shown that, in domain-specific tasks, smaller-scale models

TABLE III
KEY ENABLING TECHNOLOGIES OF ARTIFICIAL INTELLIGENCE OF THINGS FOR AGENTIC AI

Technical Domain	Key Technologies	Core Advantages	Challenges	Role in Agentic Loop
Edge Intelligence	Model Quantization, Knowledge Distillation, Split Computing, Federated Learning [27]–[29]	Reduced latency, enhanced privacy, reduced bandwidth dependency	Heterogeneity of devices, accuracy loss under resource constraints	Perception & Local Inference
Communication & Coordination	5G/6G, Semantic Communication, MARL [30]–[32]	Ultra-low latency control, intent-driven efficient transmission	Connection stability in dynamic topologies, semantic ambiguity	Collaboration & Interaction
Security & Trust	Blockchain, Trusted Execution Environment (TEE), Zero Trust Architecture [33], [34]	Immutable data, auditable operations, identity authentication	On-chain computational overhead, real-time vs. security trade-off	Guardrails & Accountability
Digital Twins	High-fidelity Physical Simulation, Sim-to-Real Transfer [35]	Low-cost trial-and-error, accelerated policy training	The “Reality Gap”, computational burden of real-time synchronization	Training & Look-ahead Planning

can achieve inference capabilities comparable to large models through fine-tuning with high-quality data, providing cost-effective solutions for vertical AIoT applications [24].

Beyond compressing the model itself, Edge-Cloud Collaborative Inference addresses the limitations of single-device computational power through a split computing architecture [25]. In this framework, different layers of the Agentic AI model are dynamically distributed across the Device-Edge-Cloud three-tier hierarchy. The device side primarily handles sensing and shallow feature extraction or processes immediate reflexive decisions, while the cloud is responsible for complex deep reasoning and long-term planning tasks. This division of labor maximizes the system’s overall inference accuracy and ensures response speed.

Furthermore, to adapt to dynamic environments, agent systems extensively use Federated Learning for distributed adaptation. This approach allows IoT agents located in different physical locations to collaboratively update a global model without sharing raw private data [26]. This enables agents to learn from collective experience and improve generalization capabilities, while also effectively reducing privacy leakage risks and bandwidth pressure associated with data transmission.

While Edge Intelligence addresses the computational and latency challenges of agent deployment, the robust operation of agent systems requires multi-dimensional technical support. To clearly define the roles of various technical modules within the Agentic AIoT architecture, this paper establishes a classification system for Key Enabling Technologies. As shown in Table III, edge computing, communication mechanisms, security architectures, and simulation platforms are outlined, highlighting the evolutionary paths of these traditional IoT technologies as they adapt to agentic requirements and their key roles within the agentic loop.

B. Communication and Coordination Mechanisms

In Agentic AIoT, communication is no longer limited to the simple transmission of raw sensor data; instead, it involves the exchange of Intents, Knowledge, and Policies among agents.

Semantic Communication represents a significant paradigm shift in communication [36]. Unlike traditional Shannon communication theory, which prioritizes the accurate transmission of bits, agent-oriented communication focuses on the accurate delivery of semantics. By leveraging the encoding capabilities of Generative AI, the transmitting agent extracts the semantic features of the information, while the receiver reconstructs it. This approach greatly enhances communication efficiency in bandwidth-constrained environments, enabling agents to convey complex task instructions with minimal data. It is particularly suitable for field or industrial sites with unstable network conditions.

In scenarios with multiple devices coexisting, Multi-Agent Collaboration and Swarm Intelligence are essential. When hundreds or thousands of AIoT agents must work together, MARL offers a decentralized coordination mechanism. By observing their local environments and the states of neighboring agents, they learn to achieve global goals without a central controller. This marks a fundamental shift in system control logic from centralized models to distributed Swarm Intelligence. Next-generation network (5G/6G) technologies provide the infrastructure necessary for the aforementioned mechanisms, particularly through the introduction of Integrated Sensing and Communication (ISAC) technology [37]. ISAC combines communication and sensing functionalities, transforming the network into a massive sensor. This is crucial for mobile agents, allowing them to use radio waves to perceive the surrounding physical environment with high precision while simultaneously transmitting data. This enables real-time, all-weather input data for highly dynamic autonomous decision-making.

C. Trust, Security, and Privacy Enablers

As AIoT systems evolve from assistive tools to autonomous agents, erroneous decisions can cause damage in the physical world. Therefore, establishing a trustworthy AIoT architecture is essential for the large-scale deployment of agents. Blockchain and smart contract technologies provide effective solutions for addressing trust issues among machines [38].

Blockchain offers agents an immutable Ledger of Actions. In multi-party collaborative scenarios, such as autonomous supply chains or smart grids, smart contracts can automatically verify whether an agent's decisions comply with predefined rules and record every interaction. This establishes the technical foundation for Machine Accountability, making agent behaviors traceable and auditable.

In the field of data privacy protection, Privacy-Preserving Computing technologies are particularly important. Agents often handle sensitive data such as camera feeds or voice recordings. The TEE provides hardware-level isolation for agent code execution, preventing malware from stealing runtime data. Additionally, Homomorphic Encryption enables agents to perform computation and planning directly on ciphertext, ensuring that data remains encrypted throughout the entire processing lifecycle and effectively eliminating the risk of data leakage. At the same time, the physical security of the system depends on robustness and adversarial defense mechanisms. Adversarial attacks targeting AI models, such as placing stickers on traffic signs to mislead autonomous vehicles, pose serious hazards in the physical world. Key technologies in this area include developing robust perception models and runtime anomaly detection mechanisms. These ensure that agents maintain safe behavior even when faced with malicious interference or sensor noise, preventing physical accidents caused by perception spoofing.

D. Digital Twins and Simulation Platforms

Since training agents directly in the physical world is prohibitively expensive and risky, Digital Twins have become the preferred training environments and verification platforms for Agentic AI. High-fidelity simulation and Sim-to-Real transfer technologies address the scarcity of training data [39]. By leveraging physics engines to construct high-fidelity virtual environments, agents can undergo millions of trial-and-error training iterations. The key challenge in this domain is bridging the Sim-to-Real Gap, ensuring that policies learned in simulation can be seamlessly transferred to real physical environments with inherent uncertainties. Currently, Domain Randomization and Online System Identification have emerged as the core technical approaches for addressing this issue.

Furthermore, Digital Twins support bidirectional synchronization and predictive control [40]. A Digital Twin is not merely a static model; it serves as a dynamic mirror synchronized in real time with the physical entity. Agents can use the twin for Look-ahead Simulation. Before executing any physical action, the agent simulates the consequences in the virtual space, selecting the optimal and safest course of action. This predictive control is particularly critical in scenarios with extremely low error tolerance, such as industrial control, surgical robots, and complex infrastructure management, effectively averting catastrophic consequences.

V. APPLICATION DOMAINS

The deep fusion of agentic AI and AIoT is transforming multiple vertical industries in the physical world. Traditional

IoT applications are generally limited to data collection and rule-based automation, but the introduction of agentic AI enables these systems to become intelligent entities that can understand environmental context, develop complex plans, and autonomously execute tasks. This section examines the transformative applications of the agentic AIoT architecture in four core domains: smart cities, industrial manufacturing, healthcare, and smart energy. As shown in Fig. 4, this application mapping diagram visually illustrates how the core capabilities of agentic AI are implemented in specific scenarios across different vertical domains, driving the evolution of various industries from automation to autonomy.

A. Smart Cities and Infrastructure

The introduction of agentic AI marks a step forward in smart city development from traditional automation toward urban autonomy. Traditional smart city technologies mainly focus on improving operational efficiency through human control, whereas agentic AI enables urban systems to independently set goals, execute tasks, and make strategic adjustments [41]. Leveraging the sensing capabilities enhanced by large language models, agent systems can not only passively respond to instructions but also exhibit partial decision-making autonomy akin to an urban brain, such as dynamically reprioritizing governance goals to cope with emergencies. This evolution requires people to look beyond mere technical deployment and re-evaluate regulatory and ethical frameworks to adapt to this urban ecosystem capable of autonomous action.

In the realm of transportation and infrastructure management, AIoT significantly enhances service efficiency and user experience through distributed collaboration. Taking autonomous driving and shared mobility scenarios as examples, agents deployed in vehicles and roadside devices can utilize graph-structured data to perceive the real-time status of urban traffic operations [42]. By adopting advanced distributed collaboration mechanisms, these agents can accurately predict passenger demand across regions and autonomously dispatch fleets without compromising passenger travel privacy. This application not only resolves resource allocation challenges in large-scale IoT environments but also achieves smart traffic flow management that is more efficient and possesses better generalization capability than traditional cloud scheduling, achieved through immediate response at the edge.

Furthermore, the application scope of agentic AI has extended to every corner of urban safety and construction, including intelligent disaster management, environmental monitoring, and public safety surveillance [43]. In these scenarios, agents distributed across drones, cameras, and sensors constitute the city's sensing and immune system. They utilize collaborative mechanisms such as federated learning to break down data barriers between different urban departments like healthcare, firefighting, and transportation, achieving cross-domain knowledge sharing while protecting sensitive information from leaking. This means that when a fire or flood occurs, environmental monitoring agents can autonomously coordinate with traffic redirection agents to plan rescue routes in real time and warn of potential risks, thereby building a rapidly responsive and highly trustworthy smart city safety net.

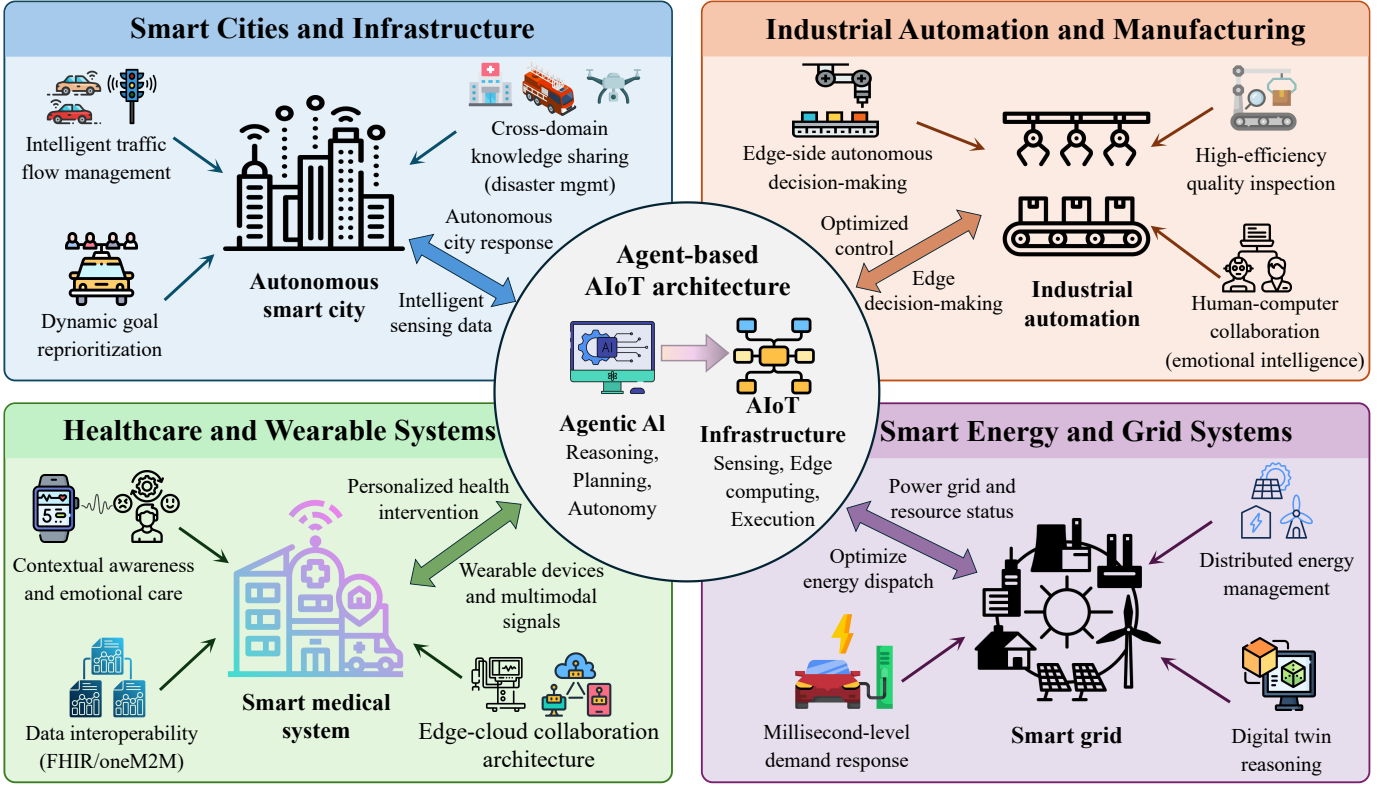


Fig. 4. Cross-domain application diagram of AIoT based on agentic AI. The central hub displays a general agent architecture integrating reasoning, planning, and autonomous functions, which supports key application scenarios such as smart cities, industrial automation, healthcare, and smart energy systems.

B. Industrial Automation and Manufacturing

In the field of industrial automation, the application of agentic AI is driving manufacturing to shift from cloud-based centralized control to edge-side autonomous decision-making. In complex real-world production lines such as precision consumable manufacturing, agent systems deployed at the IoT edge already possess the capability to perform advanced machine learning inference [44]. These edge agents no longer rely on unstable wireless bandwidth or cumbersome cloud services, but can make real-time decisions directly on-site based on sensor data. This architecture enables industrial systems to solve heterogeneous and complex quality control problems at the network edge, where agents can autonomously identify product defects and immediately adjust production parameters. Experiences drawn from these real-world deployments indicate that decentralized agentic AI systems possess significant advantages in improving inspection efficiency and reducing reliance on centralized infrastructure.

Beyond single-stage quality control, the deep fusion of agentic AI and AIoT is reshaping the sustainability of the entire manufacturing cycle, driving Industry 4.0 toward a more efficient stage. Autonomous IoT nodes, by continuously monitoring production processes and analyzing big data, can proactively identify precursors to equipment failure and perform predictive maintenance, thereby minimizing downtime and resource waste [45]. These intelligent agents take sustainable manufacturing as their core goal, helping manufacturers detect and prevent defects early through automated process optimiza-

tion and smarter decision-making. This not only improves productivity but also significantly reduces operational costs and energy consumption, enabling manufacturing enterprises to achieve synchronous green and efficient development amidst the wave of digital transformation.

As the Industrial Internet of Things (IIoT) moves toward the Industry 5.0 era, the application focus of agentic AI begins to shift from pure machine efficiency to human-centric human-machine interaction. To achieve safe and natural human-machine collaboration in noisy industrial environments, Paikrao et al. developed agent systems equipped with emotional intelligence [46]. These agents utilize advanced convolutional neural networks and signal enhancement techniques to accurately recognize operators' voice commands and their emotional states in high-noise backgrounds. By integrating this environmental perception and emotional understanding capability, industrial agents can dynamically adjust their behavior patterns based on the worker's status. Such agent systems with intelligent perception not only enhance the reliability of interaction but also provide critical technical support for constructing an adaptive and safe collaborative manufacturing environment.

C. Healthcare and Wearable Systems

In the field of healthcare, the fusion of AIoT and agentic AI is driving personalized care to evolve from simple physiological metric recording toward context-aware emotional and mental health management. Emotion, as a core driver of human

behavior and decision-making, has become a new focus for intelligent medical systems [47]. Intelligent sensing nodes deployed on wearable devices and mobile terminals are no longer merely passive collectors of data but have evolved into affective agents capable of understanding user psychological states. These agents utilize advanced deep learning algorithms such as convolutional neural networks and Transformers to fuse multimodal signals in real time, including facial expressions, voice intonation, as well as EEG and ECG. Empowered by edge computing and federated learning technologies, these agents can achieve continuous and non-invasive emotional state recognition locally, thereby providing immediate and privacy-preserving personalized interventions for depression monitoring or stress management, constructing a mental health guardianship system with high context awareness.

Addressing the challenges of chronic disease management and medical resource allocation, the agent-based edge-cloud collaborative architecture demonstrates immense potential in early disease detection and system optimization. Taking the HealthAloT architecture as an example [48], the system integrates a series of intelligent agent modules with specific functions, including a predictor agent for disease risk assessment and a cloud scheduler agent for resource management. In an empirical study of diabetes detection, this multilayer perceptron-driven agent system not only achieved high-precision risk prediction on unseen patient data, thereby supporting targeted preventive intervention, but the scheduler agent also autonomously managed requests from massive IoT devices, significantly optimizing energy consumption and latency of computational resources. This mechanism proves that agentic AI can serve not only as an assistive diagnostic tool for doctors but also as an autonomous manager of medical infrastructure, ensuring environmental sustainability of the system while improving diagnostic accuracy.

However, constructing truly comprehensive and proactive agent-based medical services must break down data silos between devices from different vendors. Existing medical AIoT development is often constrained by the incompatibility of heterogeneous data formats. To address this issue, Ji et al. proposed a unified agent system architecture based on standardized protocols (such as FHIR and oneM2M) [49]. This architecture can span different platforms such as Android, iOS, and Wear OS, seamlessly integrating raw sensor data and health survey records from heterogeneous devices like smartphones and watches. By establishing this standardized data interoperability foundation, medical agents can truly realize cross-device long-term cohort monitoring, thereby providing patients with coherent and proactive daily health management services, reducing reliance on limited hospital medical resources.

D. Smart Energy and Grid Systems

Modern power systems are undergoing an evolution from automated grids to agentic AI-based autonomous grids. The core of this transformation lies in the introduction of generative AI, LLMs, and multi-agent systems, thereby endowing the power grid with cognitive capabilities transcending traditional

rule-based control [50]. In this architecture, AIoT infrastructure is no longer merely a data transmission pipeline but serves as the sensing extension of agents. Combined with digital twin technology, agentic AI can construct real-time physical replicas of the grid in virtual space to perform context-aware reasoning. This means that agents can not only monitor anomalies but also autonomously perform full-lifecycle deduction and defense; for instance, when encountering cyberattacks, security agents can directly isolate compromised nodes through the AIoT network without human intervention, demonstrating high system robustness and decision autonomy.

In the aspect of distributed energy management, agentic AI addresses the pain point where massive heterogeneous devices struggle to collaborate. Massive IoT data generated by solar energy, wind energy, and energy storage devices, previously often used merely for status display, have now become the decision basis for edge agents [51]. By deploying agent-oriented AIoT algorithms, every photovoltaic panel or transformer is transformed into an independent intelligent entity. These distributed agents can utilize real-time data to proactively optimize their own operational strategies. For instance, an energy storage agent can autonomously decide whether to store energy or supply power back to the grid based on local weather forecasts and grid load signals. This decentralized agent mechanism transforms originally passive IoT nodes into intelligent assets capable of proactively creating value.

From the macro perspective of energy sustainability, constructing an AIoT-based agent collaborative framework is key to achieving dynamic supply-demand balance. To cope with the intermittency of renewable energy, smart grids need to possess extremely strong dynamic adaptability [52]. AIoT-empowered agent systems can autonomously perform load management at the edge through demand response mechanisms. For example, in electric vehicle charging scenarios, agents can autonomously adjust charging power within milliseconds based on real-time electricity prices and carbon emission indicators, without waiting for cloud instructions. This agent-based real-time decision-making capability not only optimizes energy allocation efficiency but also effectively promotes the integration and adoption of clean energy, providing an intelligent technical path for realizing the United Nations Sustainable Development Goals.

E. Critical Analysis of Current Deployments: From Pilots to Production

While the architectures and applications described in previous sections illustrate the transformative potential of AIoT-enabled agentic systems, a closer look at real-world deployments reveals significant gaps between research prototypes and production-ready solutions.

The Pilot-to-Production Gap: Many successfully demonstrated systems operate in controlled testbeds or high-fidelity simulations. In contrast, real-world settings introduce unpredictable sensor noise, communication delays, and physical disturbances that are seldom accounted for in controlled experiments. This discrepancy often leads to performance

degradation when systems are deployed in actual operational environments [53].

Edge Autonomy Remains Limited: Despite recent progress in edge computing, many systems described as edge-native still depend heavily on cloud servers for complex reasoning, model updates, or backup decision-making. This dependence undermines the resilience that edge intelligence is intended to provide, particularly in environments with limited or unreliable connectivity. Consequently, achieving true long-term autonomy at the edge, where devices can adapt and make decisions independently for extended periods, remains an open challenge for most real-world applications [54].

Interoperability Is Still a Major Hurdle: The vision of seamless collaboration among heterogeneous agents from different vendors is hindered by fragmented standards, proprietary protocols, and incompatible data models. Most current deployments are vertical, single-vendor solutions rather than open, multi-agent ecosystems. This fragmentation limits scalability and prevents the realization of swarm intelligence across organizational boundaries [55].

These observations underscore that progress in Agentic AIoT must address not only algorithmic and architectural challenges but also the practical issues of system integration, economic viability, and real-world robustness. The open challenges discussed in the next section should be interpreted in light of these deployment realities.

VI. OPEN CHALLENGES AND FUTURE RESEARCH DIRECTIONS

The convergence of agentic AI with AIoT transforms intelligent systems from analytic observers into embodied decision makers embedded in physical infrastructures. In this setting, intelligence is inseparable from sensing fidelity, actuation limits, and network dynamics. Failures propagate beyond computation into material and societal domains. This shift exposes fundamental limitations in prevailing AI assumptions built around static evaluation, centralized optimization, and episodic control. Advancing agentic AIoT, therefore, requires new scientific foundations that treat autonomy as a sustained, resource-bound, and socially embedded process. This section distills the most critical open challenges and outlines research directions that position AIoT as the enabling substrate for trustworthy real-world autonomy. Table IV summarizes these challenges, relating AIoT-specific constraints to qualitative risk dimensions and forward-looking research pathways across representative deployment contexts.

A. Safety, Robustness, and Trustworthiness

Safety in agentic AIoT systems arises from the interaction between learning, control, and physical execution rather than from any single algorithmic component [56]. How can safety be ensured when autonomous agents continuously sense, decide, and act through heterogeneous AIoT devices operating under uncertainty? Unlike conventional AI systems evaluated through bounded tasks, agentic systems accumulate decisions over time, allowing minor perceptual or reasoning errors to amplify through feedback loops that directly affect physical

processes [57]. Existing safety frameworks, which rely on offline testing and predefined failure modes, are poorly suited to capture this temporal compounding of risk.

Robustness and trustworthiness expose additional unresolved questions that are tightly coupled to the AIoT substrate. Agentic systems must remain reliable despite sensor drift, environmental changes, and adversarial interference. Prevailing learning paradigms assume stable data distributions and passive deployment [58], [59]. How can an agent detect that its sensing or world model has become unreliable, and how should it regulate autonomy when confidence degrades? Trust further complicates this landscape, as it emerges through sustained interaction with humans and other agents rather than from isolated system properties. How can trust be calibrated when behavior evolves through learning and coordination across the edge and cloud? [34].

Addressing these challenges requires a shift from static validation toward runtime assurance grounded in AIoT control loops. Future research should integrate uncertainty-aware reasoning, continuous self-monitoring, and adaptive control directly into decision pipelines, allowing agents to modulate autonomy based on sensing reliability and execution risk [60]. Digital twin-supported look-ahead checks, coupled with secure action logging at gateways, offer a practical pathway for enforcing safety envelopes in applications such as traffic control, industrial automation, and grid protection [61].

B. Resource-aware and Energy-efficient Autonomy

Resource constraints fundamentally shape how autonomy can be achieved in AIoT environments. Agentic systems embedded in physical infrastructures must operate continuously on energy-limited devices while supporting ongoing sensing, reasoning, communication, and learning. How can autonomous intelligence be maintained when each additional unit of reasoning incurs tangible costs in energy, latency, and carbon footprint? [62]. Approaches that focus only on model compression or hardware acceleration do not address the deeper tension between persistent autonomy and limited physical resources. This challenge is heightened by the mismatch between current AI design assumptions and embodied deployment. Most learning paradigms prioritize accuracy without considering the cumulative energy cost of long-term operation. How should an agent determine when deeper reasoning is necessary and when reactive control is sufficient? How can computation, communication, and offloading be dynamically balanced as network conditions, task urgency, and device state change? These questions highlight the lack of principled frameworks for intelligence modulation in AIoT systems.

Future research should reconceptualize autonomy as a resource-bound process that explicitly accounts for cost. Energy-aware planning frameworks are needed, where sensing rates, reasoning depth, and coordination intensity are adjusted based on contextual value and resource availability [63]. Co-design across models, hardware, and networking is equally critical, enabling split inference and control policies that reflect real device and network constraints. For example, a health monitoring agent should escalate sensing and edge inference

TABLE IV
OPEN CHALLENGES IN AIoT FOR AGENTIC AI AND THEIR MAPPING TO EMBODIED AUTONOMY CONSTRAINTS AND RESEARCH PATHWAYS

Theme	Open Challenge	AIoT Constraint Shaping Agentic AI Behavior	ES	ARG	FPR	CAF	Future Directions	Indicative Deployment Context
Safety, Robustness, Trust	How can safety be ensured under long-horizon feedback and nonstationary environments?	Closed-loop coupling across sensing-decision-actuation amplifies small errors over time	●	●	●	○	Runtime assurance with safety envelopes, twin-coupled look-ahead validation, conservative fallback control	Smart intersections, robotic cells
	How can agents detect and respond to drift, faults, or spoofing before harm occurs?	Sensor aging, distribution shift, and adversarial perturbations degrade observability and confidence	●	●	●	○	Online drift and anomaly detection at gateways, confidence-aware autonomy modulation, robust sensing fusion	Industrial monitoring, grid substations
	How can trust be calibrated when behavior evolves across edge and cloud infrastructures?	Distributed decision pathways reduce transparency and complicate attribution of actions	○	●	○	●	Provenance-linked action logs, verifiable policy versioning, intent tracing across agent interactions	Smart city operations, healthcare systems
Resource and Energy	How can autonomy be sustained under persistent energy and carbon constraints?	Battery limits and bandwidth variability make continuous autonomy costly and fragile	○	○	○	○	Event-driven sensing, adaptive cognition depth, energy-aware policy scheduling, carbon-aware offloading	Wearables, remote infrastructure
	How should computation and communication be balanced dynamically across edge and cloud?	Latency variability and heterogeneous edge capabilities destabilize split execution	○	○	○	○	Adaptive split inference, edge-first reflex loops, cost-aware coordination policies	Mobile robots, field deployments
Ethical, Legal, Societal	Who bears responsibility when harm emerges from distributed and evolving autonomy?	Control authority spans devices, gateways, cloud agents, and human overrides	○	●	○	●	Identity-bound actuation control, auditable decision trails, governance-aware rollback mechanisms	Utilities, public infrastructure
	How can fairness, consent, and transparency be preserved under pervasive sensing?	Continuous sensing and adaptive allocation decisions embed bias and reduce consent	○	●	○	●	Constraint-aware planning for equity, continuous auditing, privacy-preserving edge analytics	Public spaces, hospitals
Benchmarking, Standards	How should long-horizon autonomy be evaluated beyond static tasks?	AIoT dynamics, drift, and actuation constraints undermine short-term evaluation	○	●	●	○	Scenario-driven long-duration benchmarks, fault and drift injection, risk-aware metrics	City testbeds, factories
	How can heterogeneous AIoT systems be compared fairly and reproducibly?	Sensor fidelity, latency regimes, and actuation limits confound comparability	○	●	●	○	Standardized reporting templates, interoperable agent-device interfaces, sim-to-real gap disclosure	Cross-lab evaluation
<i>Metric definitions:</i> Embodiment Stress (ES): It measures how strongly physical interaction and actuation irreversibility amplify risk, Autonomy Readiness Gap (ARG): It measures the distance from safe sustained autonomy under open-world uncertainty, Failure Propagation Risk (FPR): It measures the likelihood that local errors cascade across AIoT layers, Control Authority Fragmentation (CAF): It measures how distributed and overlapping control rights are across devices, gateways, cloud agents, and human oversight. <i>Symbol meaning:</i> ○ low, ○ medium or context-dependent, ● high or critical.								

only during detected anomalies, while an industrial agent should remain locally autonomous during stable operation and invoke cloud-level reasoning selectively during faults. Such approaches are essential for scaling agentic AIoT sustainably.

C. Ethical, Legal, and Societal Challenges

Agentic AIoT systems introduce new forms of delegated agency that challenge existing ethical and legal frameworks. Unlike automated systems under direct human supervision, agentic systems initiate actions, coordinate with peers, and adapt policies over time. Who is responsible when harm results from behavior that emerges across distributed agents and evolving models? [34]. Traditional notions of intent and control struggle to address such distributed autonomy. These challenges become especially acute in public and socially sensitive domains where AIoT systems shape access to services

and risk exposure. Autonomous agents in cities, healthcare, and energy infrastructures increasingly influence allocation decisions with significant societal consequences. How can fairness be ensured when optimization objectives encode implicit biases or prioritize efficiency over equity? How can transparency and consent be maintained when adaptive systems make continuous decisions and their internal logic evolves beyond the original specifications?

Addressing these issues requires embedding governance into AIoT system design rather than treating ethics as an external constraint. Future research should focus on technical mechanisms that support accountability through traceable decision pathways, enforceable operational boundaries, and auditable action histories anchored in trusted infrastructure [59]. Meaningful human oversight must be designed as an operational control mode, enabling escalation and intervention within real-

time constraints. Such socio-technical foundations are critical for ensuring that agentic AIoT systems enhance collective welfare without undermining trust or social legitimacy.

D. Benchmarking, Evaluation, and Standardization

Evaluation practices must evolve to reflect the dynamic nature of agentic AIoT systems. Existing benchmarks, inherited from static machine learning, assess performance using fixed datasets and short-term tasks [64]. How should autonomy be evaluated when behavior adapts over time, interacts with other agents, and responds to environmental feedback? Current evaluation methods fail to capture long-horizon risk, robustness under drift, and sustainability of operation. The absence of standardized evaluation environments further limits scientific progress. Agentic AIoT systems operate across heterogeneous devices, sensors, and networks, making reproducibility and fair comparison difficult. How can evaluation frameworks account for variability in sensing fidelity, communication latency, and actuation capability without oversimplifying autonomy? Metrics based solely on task success obscure critical dimensions such as energy consumption, safety margins, and societal impact.

Future research should establish scenario-driven, long-duration benchmarks grounded in AIoT realities, including sensor faults, network intermittency, and constrained actuation. Digital twin-based testbeds provide a promising foundation, particularly when coupled with standardized reporting of sim to real discrepancies [65]. International standardization efforts are also needed to define shared autonomy levels, safety reporting protocols, and interoperability interfaces across devices and agents.

VII. LESSONS LEARNED AND FUTURE VISION

The integration of AIoT with agentic AI provides important lessons for researchers and practitioners. First, autonomy is not just a property of AI models but emerges from the entire system, including sensors, networks, and actuators. This is reinforced by repeated observations that even advanced agents fail when deployed on real IoT platforms with imperfect sensing and unreliable communication. Second, capability alone does not ensure safety or reliability. Systems that perform well in testing often show brittleness when confronted with new situations or adversarial conditions in the field. Third, in distributed and multi-agent settings, trust relies less on model transparency and more on the ability to audit actions and assign accountability across devices, edge nodes, and cloud services. These lessons underscore the need for a systems-level approach to designing, evaluating, and deploying autonomous AIoT systems.

Looking ahead, the most significant shift may be from isolated agents to ambient and collective intelligence. In this paradigm, intelligence is embedded in environments as a fabric of sensing, communication, and actuation that supports many interacting agents rather than a single monolithic controller. Ambient intelligence redefines the environment as an active participant, with networks sensing and inferring at the edge, digital twins offering a shared predictive foundation, and

agents negotiating intents within resource and safety constraints. Collective intelligence builds on this through coordinated multi-agent behavior, managing city-scale mobility, distributed energy, or hospital workflows with ensembles of specialized agents. They share partial beliefs, exchange semantics, and align local actions with global objectives. Achieving this vision requires scientific progress in semantic coordination, distributed learning under nonstationarity, and governance-aware control. This maintains human oversight without reducing responsiveness. If these foundations mature, AIoT for agentic AI can progress from large-scale automation to autonomy with accountability, enabling systems that are not only intelligent but also reliable, sustainable, and socially legitimate in the physical world.

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